

Paperclip: Architectural Comparison

Date: October 2026

Forked Repository vs. Official Upstream (Master Branch)

Baseline Divergence: 2,770 Commits Total

+379 FORK AHEAD COMMITS	+2,391 UPSTREAM AHEAD COMMITS	1,286 TOTAL FILES DIVERGED	0072 COMMON SCHEMA BASE
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1. Executive Summary & Strategic Positioning

The repositories diverged on **April 27, 2026** at commit `d0bdbbe11a`. Since then, the fork and upstream have developed under two distinct product visions:

The User Fork: "The Executive Desktop Cockpit" Primary Goal: Personal operating cockpit for an executive/founder managing multiple ventures from a local Windows workstation. Native Desktop Runtime: Custom Rust system tray app, desktop toast notifications, and process-tree guards. HQ Portfolio Operations: Cross-company directives, unified executive brief, aggregate receipts, and private personal spaces. Unified Communications: 3-pane email client (IMAP + Help Scout), auto-triage, AI drafting, and task creation. Clippy Assistant & Trust Loop: In-app drawer with interactive permission cards & outbound mutating tool gates. Cost Optimization: Switchboard multi-account rotation and local models (Ollama, Aider).	Official Upstream: "The Enterprise Agent Cloud" Primary Goal: Scalable, multi-tenant enterprise agent execution platform for distributed teams and cloud infrastructure. Decoupled Runner Daemon: Standalone <code>paperclip-runner</code> for remote worker nodes and Docker/Daytona sandboxes. Enterprise Governance: MCP Tool Gateway proxy, fine-grained OAuth grants, and low-trust agent isolation. Pipelines & Cases: Multi-stage DAG blocker state machines and collaborative incident desk. Skill Studio: In-browser IDE for authoring skills with evaluation kernels and automated testing. Multi-User IAM: Responsible-user commit attribution, user-specific secrets, and 30+ language translations.
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2. Deep-Dive: What Your Fork Has Built

Your fork implements several major milestones that upstream currently lists on its public roadmap as *unstarted* (🟪) or *experimental* (🟡):

Subsystem	Implementation & Capabilities	Key Repository Artifacts
Native Windows Launcher	Custom Rust executable (<code>paperclip.exe</code>) running in Windows system tray. Manages Windows named mutex single-instance locks, loopback health polling, blank UI self-healing, clean recursive process tree termination (killing tsx/postgres wrapper chains), and native toast notifications.	<code>tools/paperclip-launcher/src/main.rs</code> <code>tools/paperclip-launcher/Cargo.toml</code> <code>rebuild-paperclip.bat</code>
HQ Portfolio Directives	Root portfolio company model allowing multi-company governance. Includes one-tap directive broadcasting across companies, Portfolio Brief, Portfolio Issues, Portfolio Costs, and aggregated "Everything" console.	<code>ui/src/pages/PortfolioDirectives.tsx</code> <code>ui/src/pages/PortfolioBrief.tsx</code> <code>server/src/services/portfolio-directive.ts</code>
Integrated 3-Pane Email & Help Scout	Full mail client supporting standard IMAP and Help Scout mailboxes. Features auto-triage rules saved in DB, group-by-sender, bulk triage, steerable AI drafting popouts, and one-click "Convert to issue / hand off to agent".	<code>ui/src/pages/Email.tsx</code> (3,666 lines) <code>ui/src/pages/HelpScoutEmailView.tsx</code> <code>server/src/services/email-service.ts</code>
Clippy Assistant & Trust Gate	Persistent desktop assistant drawer. Features the "Trust Loop" mutating tool gate where destructive agent tool calls are held in draft state until approved via plain-language permission cards. Supports image/text turns and live page context injection.	<code>ui/src/pages/Clippy.tsx</code> <code>ui/src/components/ClippyDrawer.tsx</code> <code>server/src/services/tool-draft-gate.ts</code>
Switchboard Multi-Account Router	Automatic vendor account lane rotation and token management. Detects rate limits or exhausted quotas on Claude/OpenAI accounts and automatically fails over to standby credentials. Includes live provider model availability discovery.	<code>server/src/services/switchboard.ts</code> <code>server/src/services/active-account.ts</code> <code>ui/src/components/ModelPicker.tsx</code>
Calendar & Reminders	Native calendar scheduling scheduled routines and recurring reminders (daily, weekly, yearly) with desktop toast delivery.	<code>ui/src/pages/Calendar.tsx</code> <code>server/src/services/calendar.ts</code>
Ahead-of-Roadmap Systems	Memories system (<code>0081_memories.sql</code>) and Work Queues (<code>0082_work_queues.sql</code>), plus local adapters for Ollama and Aider.	<code>ui/src/pages/Memories.tsx</code> <code>ui/src/pages/WorkQueues.tsx</code> <code>packages/adapters/aider-local</code>

3. Deep-Dive: What Upstream Has Built

Upstream's 2,391 commits delivered major distributed cloud architectures and enterprise controls:

Subsystem	Implementation & Capabilities	Key Repository Artifacts
Decoupled Runner Daemon	Task execution is extracted into a standalone package (<code>paperclip-runner</code>). Enables running agents inside Docker containers, Daytona cloud sandbox leases, and Kubernetes pods with 256 MiB journal streaming.	<code>packages/paperclip-runner</code> <code>.env.runner-e2e.example</code> <code>0220_execution_workspace_runtime_leases.sql</code>
Governed MCP Tool Gateway	Server-authoritative proxy for Model Context Protocol servers. Enforces per-action policies (Allowed, Ask first, Off) with short-lived session OAuth tokens and audit logging across Asana, Sheets, Railway, and Notion.	<code>ui/src/pages/apps/gateways</code> <code>0151_tool_gateway_sessions.sql</code> <code>0159_named_mcp_gateways.sql</code>
Pipelines & Case Incident Desk	Multi-stage DAG state machines with automatic transitions, blocker dependencies, and upstream drift tracking. Cases desk provides a collaborative environment for incident investigations and customer issue triage.	<code>ui/src/pages/Pipelines.tsx</code> <code>ui/src/pages/Cases.tsx</code> <code>0113_pipeline_foundation.sql</code>
Skill Studio & Evals	In-browser IDE for authoring agent skills. Includes evaluation kernels (<code>paperclip-eval-kernel</code>), regression testing against saved test fixtures, version history rollback, and a shared Skills Store catalog.	<code>ui/src/pages/SkillStudio.tsx</code> <code>packages/paperclip-eval-kernel</code> <code>0099_skills_store_foundation.sql</code>
Enterprise IAM & Attribution	Multi-user identity management where git commits and external connections are attributed to the specific human instructing the work ("Responsible User"). Includes user-specific encrypted secrets and low-trust directory sandboxing.	<code>0128_user_specific_secrets.sql</code> <code>0129_agent_api_key_responsible_user.sql</code> <code>0097_low_trust_source_trust.sql</code>
Git Worktrees & Concurrency	First-class support for git worktrees, enabling multiple agents to develop and test concurrently on isolated branches within the same repository without lock collisions.	<code>packages/server/src/services/worktrees</code> <code>0220_execution_workspace_runtime_leases.sql</code>
Global Internationalization	Full i18n translation coverage supporting 30+ languages, PWA controls, and browser-use cloud preview integrations.	<code>ui/src/i18n/locales/*.json</code> <code>0290_browser_use_cloud.sql</code>

4. Database Schema Divergence Analysis

Critical Architectural Note: Both codebases branched after migration `0072_large_sandman.sql`.

Because migration sequence numbers were assigned independently on both branches:

- **Fork:** Created migrations `0073_plugin_local_source_path.sql` through `0100_backfill_missing_issue_identifiers.sql`.
- **Upstream:** Created migrations `0073_shiny_salo.sql` through `0294_chilly_marvel_apes.sql` (over 220 new migrations).

Impact: A standard `git merge` or `git rebase` will cause severe, irreconcilable database schema collisions and unresolvable migration sequence state. Any feature integration must be done modularly.

5. "Which One Is Better?" Decision Matrix

User Profile & Operating Model	Winner	Rationale
Solo Founder / Executive Operator	Your Fork	Your fork provides a complete operational desktop environment (HQ portfolio overview, integrated email/Help Scout client, Clippy trust gate, desktop toast reminders). Upstream lacks desktop and communications infrastructure.
Windows Workstation Daily Driver	Your Fork	Upstream has no desktop executable. Your Rust system tray app, process tree cleanup, and restart/update scripts make running on Windows seamless and crash-resilient.
Cost Minimization & Local LLMs	Your Fork	Switchboard enables rotating multiple accounts automatically when rate-limited. Local Ollama and Aider adapters eliminate reliance on cloud API spend.
Multi-Human Team / Organization	Upstream	Upstream offers enterprise RBAC, "Responsible User" commit attribution, per-user private secrets, and decision desk tracking.
High-Scale Concurrent Agent Fleet	Upstream	Upstream's separated runner daemon, Daytona sandbox leases, and Git worktrees allow 10+ agents to develop code concurrently without workspace collisions.
Strict Security & Compliance Audit	Upstream	Upstream provides formal MCP Tool Gateways, low-trust agent isolation, OAuth revocation, and evaluation verification suites.

6. Strategic Recommendations & Action Plan

Recommended Strategy: Treat your fork as a specialized executive distribution. Do not attempt a full git merge.

1. **Keep Your Fork as Primary:** Continue running your fork for your day-to-day operations. Its native desktop integration, email triage, and Clippy assistant cannot be replaced by upstream today.
2. **Cherry-Pick High-Value Upstream Adapters:** If you want new agent harnesses (such as `grok-local`, `kiml-local`, or `cursor-cloud`), cleanly cherry-pick or copy the adapter packages from `packages/adapters` into your fork without touching the server core.
3. **Package Key Superpowers as Plugins:** Upstream's official plugin system is now mature. Consider extracting Switchboard and Calendar into standalone Paperclip plugins that can run on any version.
4. **Monitor Upstream Runner Maturity:** If you scale to remote cloud servers or require Daytona sandbox pools, consider deploying an upstream instance as a headless runner worker while keeping your executive cockpit intact.